

ConvTranspose2d#

<https://pytorch.org/docs/stable/generated/torch.nn.quantized.ConvTranspose2d.html>

Description:

Applies a 2D transposed convolution operator over an input image composed of several input planes. For details on input arguments, parameters, and implementation see `ConvTranspose2d`. For special notes, please, see `Conv2d weight (Tensor)` – packed tensor derived from the learnable weight parameter. `scale (Tensor)` – scalar for the output scale `zero_point (Tensor)` – scalar for the output zero point

Function Signature

```
class torch.nn.quantized.ConvTranspose2d(in_channels,
out_channels, kernel_size, stride=1, padding=0,
output_padding=0, groups=1, bias=True, dilation=1,
padding_mode='zeros', device=None, dtype=None)[source]#
```

Variables

Code Examples

```
>>> # QNNPACK or FBGEMM as backend
>>> torch.backends.quantized.engine = 'qnnpack'
>>> # With square kernels and equal stride
>>> import torch.nn.quantized as nnq
>>> m = nnq.ConvTranspose2d(16, 33, 3, stride=2)
>>> # non-square kernels and unequal stride and with padding
>>> m = nnq.ConvTranspose2d(16, 33, (3, 5), stride=(2, 1),
padding=(4, 2))
>>> input = torch.randn(20, 16, 50, 100)
>>> q_input = torch.quantize_per_tensor(input, scale=1.0,
zero_point=0, dtype=torch.qint8)
>>> output = m(q_input)
>>> # exact output size can be also specified as an argument
>>> input = torch.randn(1, 16, 12, 12)
>>> q_input = torch.quantize_per_tensor(input, scale=1.0,
zero_point=0, dtype=torch.qint8)
>>> downsample = nnq.Conv2d(16, 16, 3, stride=2, padding=1)
>>> upsample = nnq.ConvTranspose2d(16, 16, 3, stride=2,
padding=1)
>>> h = downsample(q_input)
>>> h.size()
torch.Size([1, 16, 6, 6])
>>> output = upsample(h, output_size=input.size())
>>> output.size()
torch.Size([1, 16, 12, 12])
```

Detailed Documentation

Documentation

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For special notes, please, see `Conv2d`

`weight (Tensor)` – packed tensor derived from the learnable weight parameter.

`scale (Tensor)` – scalar for the output scale

`zero_point (Tensor)` – scalar for the output zero point

See `ConvTranspose2d` for other attributes.

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